AME 5101 - Applied Linear Algebra

Assignment 3A

Maximum marks: 10 Duration: 1 hour

Reading the Backpropagation Notes – A Worked Example.

Sections 1 to 4.1 of the notes [CO3-PO3-BL3]

The questions below follow the notes *Backpropagation with Cross-Entropy Loss*, Sections 1 to 4.1. Each answer first says how to think about the question, and then gives the answer.

(a) [1] A classifier has $K = 4$ classes. For an example whose true class is class 3, write down its label c and its target $\mathbf{y}$. What does row i of a weight matrix $\mathbf{W}^{(l)}$ contain?

Answer. *How to think.* The notes use two words for the same fact. The *label* is a number, the index of the true class. The *target* is a vector of length K with a single 1 at that index.

So the label is $c = 3$ and the target is $\mathbf{y} = (0,\ 0,\ 1,\ 0)^\top$.

Row i of $\mathbf{W}^{(l)}$ holds the weights going *into* unit i of layer l, one weight for each unit of layer $l-1$. That is why $z_i^{(l)}$ is row i of $\mathbf{W}^{(l)}$ times the column vector $\mathbf{a}^{(l-1)}$, plus $b_i^{(l)}$.

Marking: [½] label and target; [½] meaning of row i.

(b) [2] A fully connected network has layer sizes $3 \to 4 \to 2$. Give the shape of each weight matrix and bias vector, and count the weights, the biases and the total number of parameters.

Answer. *How to think.* $\mathbf{W}^{(l)}$ has one row per unit of layer l and one column per unit of layer $l-1$, so its shape is $n_l \times n_{l-1}$. The bias $\mathbf{b}^{(l)}$ has one entry per unit of layer l.

Layer 1 (3 inputs, 4 units): $\mathbf{W}^{(1)} \in \mathbb{R}^{4\times 3}$, so 12 weights, and $\mathbf{b}^{(1)} \in \mathbb{R}^4$, so 4 biases.

Layer 2 (4 inputs, 2 units): $\mathbf{W}^{(2)} \in \mathbb{R}^{2\times 4}$, so 8 weights, and $\mathbf{b}^{(2)} \in \mathbb{R}^2$, so 2 biases.

In all: $12 + 8 = 20$ weights and $4 + 2 = 6$ biases, so 26 parameters.

Marking: [1] shapes; [1] counts and total.

(c) [2] In Section 2, why do we take the logarithm of the likelihood, and why do we then negate it? For the target $\mathbf{y} = (0,\ 0,\ 1,\ 0)^\top$, show that $\prod_{k=1}^{4} q_k^{y_k}$ is equal to q_3.

Answer. *How to think.* Each of the four steps in Section 2 fixes one practical problem. Name the problem, then the fix.

Why the logarithm. The likelihood is a product of many probabilities, each at most 1. Such a product becomes so small that the computer stores it as 0, and its derivative by the product rule is clumsy. The logarithm turns the product into a sum. Since ln is increasing, the θ that maximises the log-likelihood also maximises the likelihood.

Why negate. Optimisers are written to *minimise*. Maximising ℓ is the same as minimising $-\ell$, the negative log-likelihood.

The product. Write out all four factors:

$$\prod_{k=1}^{4} q_k^{y_k} = q_1^0\, q_2^0\, q_3^1\, q_4^0 = 1 \cdot 1 \cdot q_3 \cdot 1 = q_3.$$

Every class with $y_k = 0$ contributes a factor 1, so only the true class is left.

Marking: [1] the two reasons; [1] the product.

(d) [1] A freshly initialised network with small random weights is trained on $K = 4$ classes. About what loss should it give on its first example, and why?

Answer. *How to think.* Small weights give logits that are all close to zero, so the softmax cannot prefer any class.

With all logits nearly equal, every $q_k \approx 1/K = 1/4$. The loss is $L = -\ln q_c \approx -\ln(1/4) = \ln 4 \approx 1.386$ nats, whatever the label. A much larger first loss usually means a bug or weights that are too large.

Marking: [½] $q_k \approx 1/4$; [½] $L \approx \ln 4$.

(e) [2] State the gradient of the loss with respect to the logit z_k. For a model output $\mathbf{q} = (0.1,\ 0.2,\ 0.6,\ 0.1)^\top$ and label $c = 3$, compute $\boldsymbol{\delta}^{(3)}$ and check that its entries sum to zero. Why must they always sum to zero?

Answer. *How to think.* Section 3 shows that the gradient at the logits is simply the prediction minus the target. Write down the target first, then subtract.

$\dfrac{\partial L}{\partial z_k} = q_k - y_k$, so $\boldsymbol{\delta}^{(3)} = \mathbf{q} - \mathbf{y}$. With $c = 3$, $\mathbf{y} = (0,\ 0,\ 1,\ 0)^\top$ and

$$\boldsymbol{\delta}^{(3)} = (0.1 - 0,\ 0.2 - 0,\ 0.6 - 1,\ 0.1 - 0)^\top = (0.1,\ 0.2,\ -0.4,\ 0.1)^\top.$$

Sum: $0.1 + 0.2 - 0.4 + 0.1 = 0$. This always happens, because $\sum_k q_k = 1$ and $\sum_k y_k = 1$, so $\sum_k (q_k - y_k) = 1 - 1 = 0$. Only the true class has a negative entry: the gradient pushes its logit up and the others down.

Marking: [½] formula; [1] $\boldsymbol{\delta}^{(3)}$ and the check; [½] reason.

(f) [2] A layer has two inputs a_1, a_2 and two units, with $z_i = W_{i1}a_1 + W_{i2}a_2 + b_i$ for $i = 1, 2$, and $\delta_i = \partial L/\partial z_i$ is known. Write out z_1 and z_2. Find $\partial L/\partial W_{21}$ and $\partial L/\partial b_2$. Arrange the four weight gradients in a matrix and show that it equals the outer product $\boldsymbol{\delta}\mathbf{a}^\top$.

Answer. *How to think.* As in Section 4.1, first write out each z_i and see which weights it contains. A weight appears in only one z_i, so only one term of the chain rule survives.

$$z_1 = W_{11}a_1 + W_{12}a_2 + b_1, \qquad z_2 = W_{21}a_1 + W_{22}a_2 + b_2.$$

W_{21} appears only in z_2, multiplied by a_1, so $\partial z_2/\partial W_{21} = a_1$ and

$$\frac{\partial L}{\partial W_{21}} = \frac{\partial L}{\partial z_2}\,\frac{\partial z_2}{\partial W_{21}} = \delta_2\, a_1.$$

b_2 appears only in z_2, as a plain added term, so $\partial z_2/\partial b_2 = 1$ and $\partial L/\partial b_2 = \delta_2$. In the same way $\partial L/\partial W_{ij} = \delta_i a_j$ for every i, j. Placing entry (i, j) in row i, column j:

$$\begin{bmatrix} \delta_1 a_1 & \delta_1 a_2 \\ \delta_2 a_1 & \delta_2 a_2 \end{bmatrix} = \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} \begin{bmatrix} a_1 & a_2 \end{bmatrix} = \boldsymbol{\delta}\mathbf{a}^\top.$$

Row i is δ_i times the input vector; column j is a_j times $\boldsymbol{\delta}$.

Marking: [½] z_1, z_2; [½] $\partial L/\partial W_{21}$; [½] $\partial L/\partial b_2$; [½] outer product.

The Assignment Question.

Q1. [CO3-PO3-BL3] **10 Marks**

Answer the following questions on the same lines as the worked example shown above. Show your reasoning.

(a) **[1]** A classifier has $K = 5$ classes. For an example whose true class is class 2, write down its label c and its target $\mathbf{y}$. What does column j of a weight matrix $\mathbf{W}^{(l)}$ contain?

(b) **[2]** A fully connected network has layer sizes $4 \to 3 \to 2$. Give the shape of each weight matrix and bias vector, and count the weights, the biases and the total number of parameters.

(c) **[2]** Explain in your own words why Section 2 takes the logarithm of the likelihood, and why it then negates it. For the target $\mathbf{y} = (0,\ 1,\ 0,\ 0,\ 0)^\top$, show that $\prod_{k=1}^{5} q_k^{y_k}$ is equal to q_2.

(d) **[1]** A freshly initialised network with small random weights is trained on $K = 5$ classes. About what loss should it give on its first example, and why?

(e) **[2]** For a model output $\mathbf{q} = (0.05,\ 0.7,\ 0.1,\ 0.1,\ 0.05)^\top$ and label $c = 2$, compute $\boldsymbol{\delta}^{(3)} = \mathbf{q} - \mathbf{y}$ and check that its entries sum to zero. Which entry is negative, and what does that tell you about how a gradient-descent step changes the logits?

(f) **[2]** A layer has three inputs a_1, a_2, a_3 and two units, with $z_i = W_{i1}a_1 + W_{i2}a_2 + W_{i3}a_3 + b_i$ for $i = 1, 2$, and $\delta_i = \partial L/\partial z_i$ is known. Write out z_1 and z_2. Find $\partial L/\partial W_{13}$ and $\partial L/\partial b_1$. Arrange the six weight gradients in a matrix and show that it equals the outer product $\boldsymbol{\delta}\mathbf{a}^\top$.